Federico Milana

Ph.D. in Human-AI Interaction with 5+ years of experience designing, implementing, and evaluating ML classifiers, LLMs and chatbots. Specialized in explainability and human-centered ML.

EXPERIENCE

Graduate Machine Learning Engineer
September 2025 – Present / Cleo
Built and evaluated LLM systems, including fine-tuned 8B models and chat routers, improving accuracy from 55%→83% and 80%→92%.

Postgraduate Teaching Assistant
January 2021 – May 2024 / University College London
Assisted in teaching Python in practical tutorials and assessed coursework for Human-Computer Interaction MSc students.

Research Assistant
August 2019 – November 2019 / University College London
Evaluated the effects of UI design factors on user trust and autonomy delegation to chatbots, contributing to recommendations for chatbot interfaces.

EDUCATION

Ph.D. in Human-AI Interaction in Machine Learning
September 2020 – November 2024 / University College London
Thesis: "Evaluating Interaction with Machine Learning Text Classifiers and Interpretability Techniques". Research on user interaction, explainability, and iterative model refinement in text classification.

MSc in Human-Computer Interaction
September 2018 – August 2019 / University College London
Graduated with Distinction, awarded Dean's List honors for top 5% academic performance.

BSc in Computer Science
September 2015 – August 2018 / King's College London
Graduated with First-class honors. Specialization in Software Engineering.

PUBLICATIONS

AI-Powered Corpus Translation Studies at Work / Springer Nature / *https://doi.org/10.1007/978-981-95-5933-6*
Applied Generative AI to digital Corpus Translation Studies, streamlining analysis of seventeenth-century translations while ensuring reliability through cross-validation and critical oversight.

Understanding Interaction with Machine Learning through a Thematic Analysis Coding Assistant: A User Study / CSCW '25 / *https://doi.org/10.1145/3711095*
Explored how an Interactive ML tool facilitates critical reflection, insights, and adaptation in analytical processes while uncovering misconceptions about ML in non-expert users.

Chatbots as Advisers: the Effects of Response Variability and Reply Suggestion Buttons / CUI '23 / *https://doi.org/10.1145/3571884.3597132*
Demonstrated how response variability, human likeness and reply suggestion buttons significantly increase chatbot advice following.

PROJECTS

QLoRa Bot
April 2025 – August 2025 / *https://github.com/fmilana/QLoRaBot*
Fine-tuned LLaMA-3-8B-Instruct on personal chats using Quantized Low-Rank Adaptation for efficient training, achieving conversational alignment with original user tone and style.

Translation Annotator
March 2025 – August 2025 / *https://github.com/fmilana/TranslationAnnotator*
Developed an Electron application to visualize manual and AI-generated annotations of literary translation regularities using OpenAI and Anthropic's APIs.

Interpretable Text Classification
November 2023 – July 2024 / *https://github.com/fmilana/explanations*
Implemented and evaluated LIME, SHAP, and transformers-interpret heatmaps for text classification using XGBoost and fine-tuned DistilBERT in user studies.

Thematic Analysis Coding Assistant
January 2021 – September 2024 / *https://github.com/fmilana/taca*
Developed a desktop application implementing XGBoost to assist the qualitative coding phase of thematic analysis, enabling iterative model refinement.

CONTACT

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https://linkedin.com/in/fmilana
https://github.com/fmilana

SKILLS

Machine Learning
NLP
LLMs
Fine-tuning
Explainability

Programming
Python
PyTorch
Hugging Face
XGBoost
scikit-learn
numpy
pandas

Research
User studies
Qualitative analysis
Academic writing

CERTIFICATIONS

Deep Learning
February 2025
DeepLearning.AI

Machine Learning
February 2025
Stanford University

Mathematics for ML
January 2025
Imperial College London

AWARDS

UCL Faculty of Brain Sciences 2018/2019 Dean's List
August 2019
Academic performance equivalent to top 5% student achievement

VOLUNTEERING

Reviewer for CHI '24, '25

Reviewer for IMWUT '24

Student Volunteer for CHI '23
Hamburg, Germany